Vance Vanvolkenburgh
655.662 — Introduction to Healthcare Systems Engineering
Module 7 | M7A3: Design Requirements

Design Requirements — Peritoneal Dialysis Remote Patient Monitoring (PD-RPM)

Design requirements constrain how PD-RPM is **realized** rather than what it does. The five below are drawn from the design-element set the module names — reliability, availability, maintainability, producibility, human factors — and each one is assessed against the design itself at Preliminary and Critical Design Review, not against a running system. Each states a single **shall**, carries a numeric criterion, and names the review or qualification test that closes it. They are deliberately disjoint from M7A1, which defines what crosses the system boundary, and from M7A2, which defines what the software does with what it receives.

Req	Design area	Applies to	Class
DR-1	Availability and fault tolerance	Monitoring and alerting path, EG-1 through NG-1	Performance
DR-2	Home operating environment and installability	Home Gateway (HG-1)	Characteristic
DR-3	Patient-facing use safety	Patient Tablet (PT-1), HG-1 indicators, patient materials	Constraint
DR-4	Field maintainability	Home Gateway (HG-1), Device Management Server (DM-1)	Characteristic
DR-5	Radio module partitioning	Home Gateway (HG-1)	Constraint

DR-1 — Availability and fault tolerance · Monitoring and alerting path, EG-1 through NG-1 · design performance

The monitoring and alerting path from the Edge Gateway (EG-1) through the Notification Gateway (NG-1) shall be designed to achieve an operational availability of not less than 0.98 measured over any rolling 90-day period, with no single hardware, service, or availability-zone failure capable of interrupting alert delivery for more than 5 minutes.

Allocated from: KPP Operational Availability ($A_o \geq 0.98$); KPP Alert Latency
Verification: Analysis by reliability block diagram with MTBF and MTTR allocated to each element, reviewed at PDR; confirmed by fault-injection testing of each single point of failure before CDR

DR-2 — Home operating environment and installability · Home Gateway (HG-1) · design characteristic

The Home Gateway (HG-1) shall be designed for unattended operation in a domestic bedroom: enclosure rated to IP32, operating across 10 to 35 °C and 10 to 85 percent relative humidity non-condensing, cooled by convection with no moving parts, emitting no more than 30 dBA measured at 1 meter, and installable by the patient or care partner without tools in not more than 10 minutes.

Allocated from: OR-2 (no operational burden beyond running the cycler); operational need ON-2
Verification: Qualification test per IEC 60601-1-11 for the home healthcare environment, acoustic measurement in an anechoic chamber, and timed out-of-box installation by 10 untrained participants

DR-3 — Patient-facing use safety · Patient Tablet (PT-1), HG-1 indicators, patient materials · design constraint

All patient-facing elements of PD-RPM shall be designed under an IEC 62366-1 usability engineering process, shall present text at not less than a 16-point equivalent with a contrast ratio of at least 4.5 to 1, and shall be validated by a summative usability study with not fewer than 15 participants per distinct user group in which zero use errors lead to a hazardous situation.

Allocated from: KPP Patient Daily Interaction Time (≤ 60 seconds); OR-3; the adult ESKD home population, which skews older and includes diabetic visual impairment
Verification: Inspection of the usability engineering file at CDR, followed by the summative study executed on production-equivalent units

DR-4 — Field maintainability · Home Gateway (HG-1), Device Management Server (DM-1) · design characteristic

The design shall permit firmware and configuration update of the Home Gateway (HG-1) from the Device Management Server (DM-1) without a home visit, applying no update during an active treatment session, reverting automatically to the previously known-good image after 2 consecutive failed boots, and achieving a fleet update success rate of not less than 98 percent within 30 days of release.

Allocated from: KPP Operational Availability; risk R-1 (silent gateway) and mitigation M-3 from M5A2
Verification: Demonstration on a 50-unit pilot fleet including deliberately corrupted packages and a mid-session update attempt, with rollback and session protection both observed

DR-5 — Radio module partitioning · Home Gateway (HG-1) · design constraint

The Home Gateway (HG-1) shall be designed with the cellular radio as a field-replaceable module on a documented internal interface, such that substituting a different certified radio requires changes to no element other than the radio abstraction layer, requires no change to the enclosure, power, or thermal design, and requires no re-certification of the host board.

Allocated from: The re-trade trigger recorded in M4A2 step 12, against carrier sunset of LTE-M service
Verification: Inspection of the module interface specification at PDR, and demonstration by fitting a second, different certified radio module to a production-equivalent host

Where the tension sits. DR-5 buys insurance against a carrier technology sunset and pays for it in unit cost: a socketed, separately certified radio is more expensive than one soldered to the host board, across every unit shipped. The M4A2 trade study selected the cellular gateway on the assumption that LTE-M remains available for the service life of the fleet, and recorded carrier sunset as the trigger to re-open that trade. DR-5 is the cheaper response to that trigger, but it is not free, and it should be re-examined at PDR against the bill-of-materials target rather than treated as settled.

Two of these requirements govern things no amount of software can repair. DR-2 constrains a device that will sit in a bedroom beside a sleeping patient every night for years, where a cooling fan is a reason to unplug it. DR-3 constrains a user population that is older than average and frequently visually impaired by diabetes, where a 15-participant summative study is the regulatory floor rather than a generous sample. Both are design decisions in the strict sense: once the enclosure is tooled and the interaction model is fixed, neither is recoverable later in the program at acceptable cost.